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Languages:
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Muhammad Nouman

PhD Researcher in Autonomous Systems

Research Summary

PhD researcher working on reliable sensing, localization, and security for connected autonomous systems, combining Bluetooth 6.0/BLE localization, machine-learning-based anomaly and attack detection, and UAV-based inspection. Research interests include learning-based methods for safety and robustness in autonomous and physical AI systems.

Education

PhD in Autonomous Systems

Politecnico di Bari, Italy, 2024 – 2027 (Expected)

Thesis: *Methodologies and Protocols for Innovation and Interoperability between Internet of Things Standards in Home Automation.*

Master of Science in Computer Science (MScS)

COMSATS University Islamabad, Islamabad Campus, Pakistan, 2020 – 2022

Thesis: *Efficient Malicious Nodes' Detection using Machine Learning and Distributed Data Storage using Blockchain and IPFS in WSN.*

Experience

Researcher

[Telematic Lab](#), Politecnico di Bari, Italy, Nov 2024 – Present

- Researching Bluetooth 6.0 advertising and localization across multiple low-power IoT standards, with three manuscripts in preparation.
- Developed a machine-learning-based histogram boosting approach for attack detection in IoT networks (IFIP WMNC 2025).
- Contributed to research on video quality assessment for UAV-based bridge inspections (IEEE DESSERT 2025).

International Research Collaboration

University of North Dakota, USA, Jan 2023 – Aug 2024

- Collaborated with [international student research groups](#) at the School of Electrical Engineering and Computer Science, working on blockchain- and ML-based security frameworks.
- Implemented experimental code, validated research findings, and contributed to two journal publications.

Research Assistant

[ComSens Research Lab](#), COMSATS University Islamabad, Pakistan, Jul 2021 – Oct 2022

- Conducted research on ML-based malicious node detection in wireless sensor networks and supervised student research projects.
- Responsibilities included producing research papers, reviewing student work, supporting idea development, and delivering lectures.



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Journal Publications

1. **M. Nouman**, U. Qasim, H. Nasir, A. Khalid, A. Almasoud, M. Imran, N. Javaid. *Malicious Node Detection using Machine Learning and Distributed Data Storage using Blockchain in WSNs*, *IEEE Access*, 2023. DOI: [10.1109/ACCESS.2023.3236983](https://doi.org/10.1109/ACCESS.2023.3236983)
2. S. Ismail, **M. Nouman**, D. W. Dawoud, H. Reza. *Towards an Integrated Security Framework Using Blockchain and Machine Learning*, *Blockchain: Research and Applications*, 2023. DOI: [10.1016/j.bcr.2023.100174](https://doi.org/10.1016/j.bcr.2023.100174)
3. S. Ismail, **M. Nouman**, H. Reza, F. Vasefi, H. K. Zadeh. *A Blockchain-based Fish Supply Chain Framework for Maintaining Fish Quality and Authenticity*, *IEEE Transactions on Services Computing*, 2024. DOI: [10.1109/TSC.2024.3422897](https://doi.org/10.1109/TSC.2024.3422897)

Conference Publications

1. **M. Nouman**, N. Cordeschi, A. Fascista. *Machine Learning-Based Histogram Boosting Approach for Attack Detection in IoT Networks*. In *16th IFIP Wireless and Mobile Networking Conference (WMNC)*, Leuven, Belgium, 2025, pp. 53–57. DOI: [10.23919/WMNC67099.2025.11299281](https://doi.org/10.23919/WMNC67099.2025.11299281)
2. O. Pohudina, N. Cordeschi, **M. Nouman**, A. Fascista. *Assessing Video Quality in Bridge Inspections with Unmanned Aerial Vehicles*. In *Proceedings of the 15th IEEE International Conference on Dependable Systems, Services and Technologies (DESSERT)*, Athens, Greece, 2025. (accepted for publication).

Manuscripts in Preparation

1. **M. Nouman**, P. Boccadoro, N. Cordeschi, A. Fascista. *Bluetooth 6.0 for IoT: A Survey on Advancements from Decision-Based Advertising Filtering to Channel Sounding*. Target: *IEEE Internet of Things Journal*.
2. **M. Nouman**, P. Boccadoro, N. Cordeschi, A. Fascista. *iBeacon Presence and Motion-State Estimation from Irregular BLE Advertisement Logs*. Target: *IoTBDs* (position paper).
3. **M. Nouman**, N. Cordeschi, A. Fascista, P. Boccadoro. *Auxiliary Channel-Selection-Aware Throughput Optimization for Bluetooth Low Energy Extended Advertising*. Target: *IEEE Wireless Communications Letters*.

Conferences & Summer Schools Attended

1. 2026 PhD Summer School on Autonomous Systems, Politecnico di Bari, Italy 7–11 September 2026.
2. Automatica.it 2025, Certificate of Attendance, Perugia and Assisi, Italy 3–5 September 2025.
3. Summer School: Shaping the Evolution of Machine Learning in the Fields of Information Engineering and Security: From Theory to Practice, Brixen, Italy 7–11 July 2025.
4. 6th International Conference on Industry 4.0 and Smart Manufacturing (ISM 2024), Certificate of Attendance, Prague, Czech Republic 20–22 November 2024.

Skills

• Research & Development • Machine Learning • Deep Learning
• Anomaly & Attack Detection • BLE Localization • IoT & Wireless Networks • Blockchain • Academic Mentorship • Technical and Research Report Writing

Programming Languages

• Python • C++ • MATLAB • Solidity • LaTeX